GENERAL ASSEMBLY

GRADIENT DESCENT

Matt Brems

DSI+

DATA SCIENCE PROCESS

1. Define the problem.
2. Obtain the data.
3. Explore the data.
4. Model with data.
5. Evaluate the model.
6. Answer the problem.

GRADIENT DESCENT

- Recall: What is a loss function?

GRADIENT DESCENT

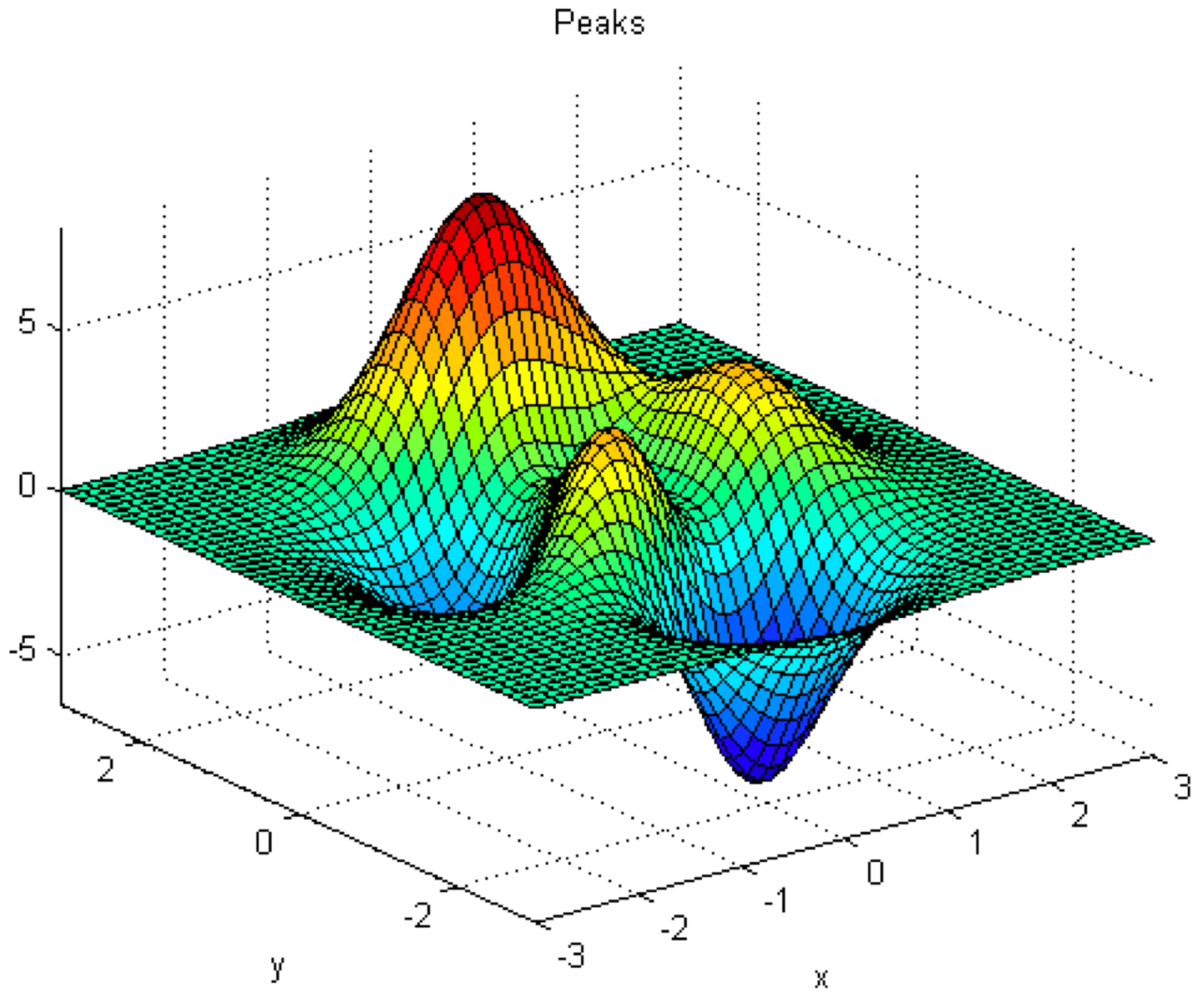
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- Recall: What are common loss functions we've used?

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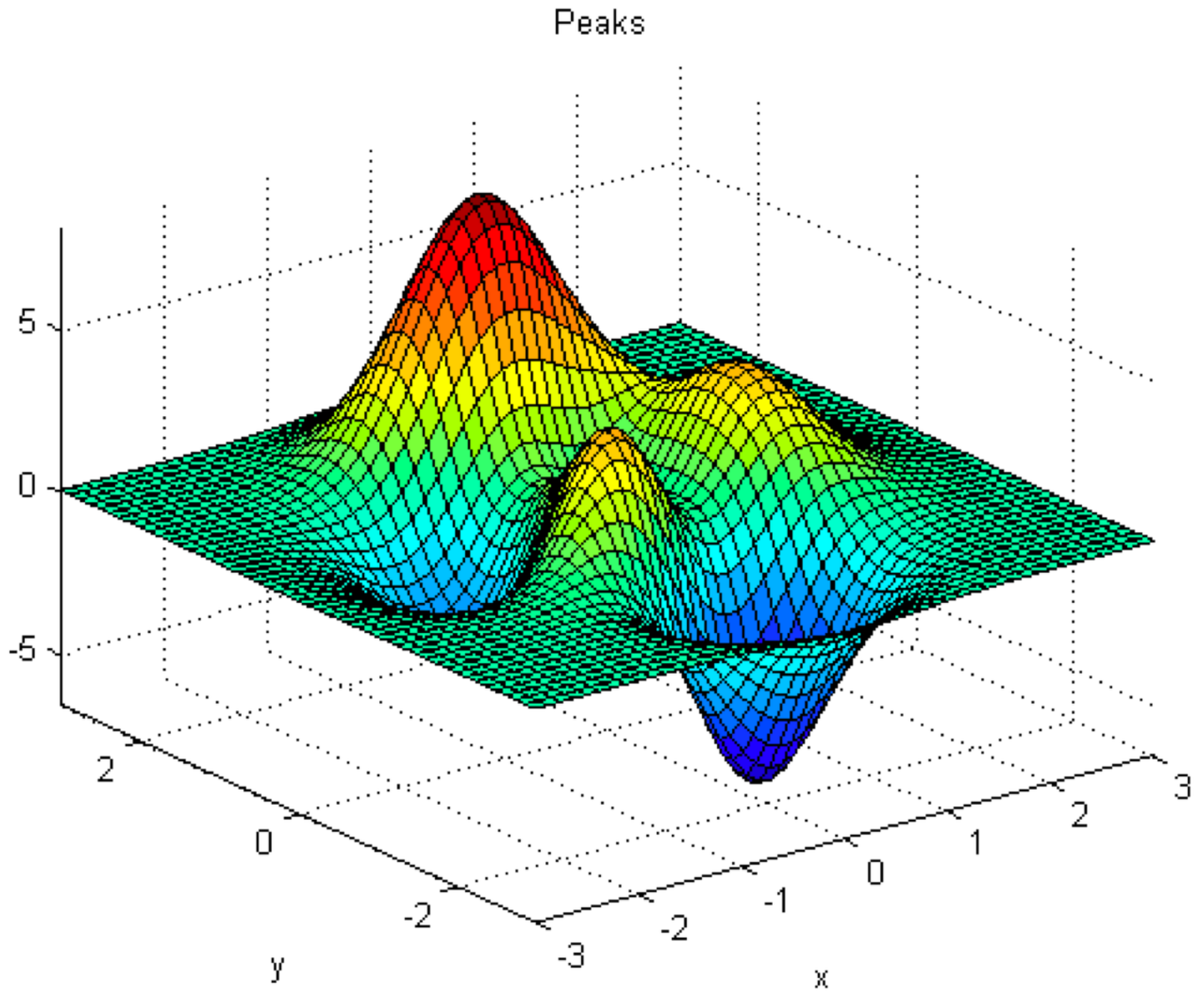
- Recall: What is a loss function?
- Recall: What are common loss functions we've used?
- Recall: What purposes do our loss function serve?

LOSS FUNCTION



- What do we do with this?

LOSS FUNCTION



- What do we do with this?
- How do we do it?

GRADIENT DESCENT

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- Algorithm Sketch:
  1. Start by making a guess for the optimal parameter value.
  2. Calculate the loss given that parameter value.
  3. Update guess to decrease loss.
  4. Keep going until loss is “sufficiently minimized.”

MATH REPRESENTATION

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- $\frac{\partial L}{\partial \beta_1}$: Gradient of Loss Function with respect to β_1.
  - Tells us direction of steepest slope. (Gravity!)

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- Keep going until $\left|\hat{\beta}_{1,i+1} - \hat{\beta}_{1,i}\right|$ is sufficiently small. (Why?)

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- Step 6: Repeat steps 3 through 5 until "stopping condition" is met.

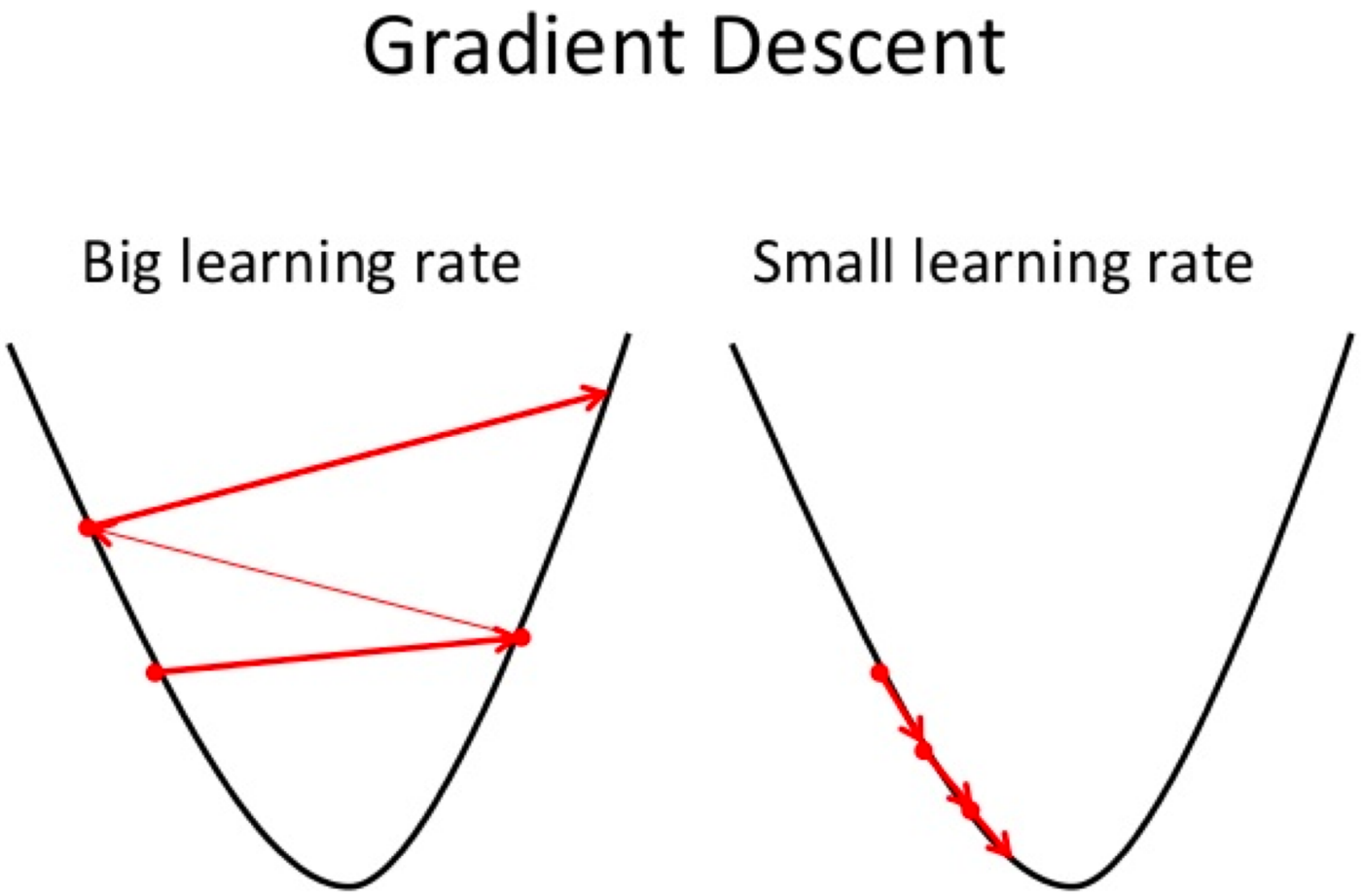
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- Step 7: $\hat{\beta}_1 = \hat{\beta}_{1,n}$, where n is the number of iterations of gradient descent.

INTUITIVE RECAP

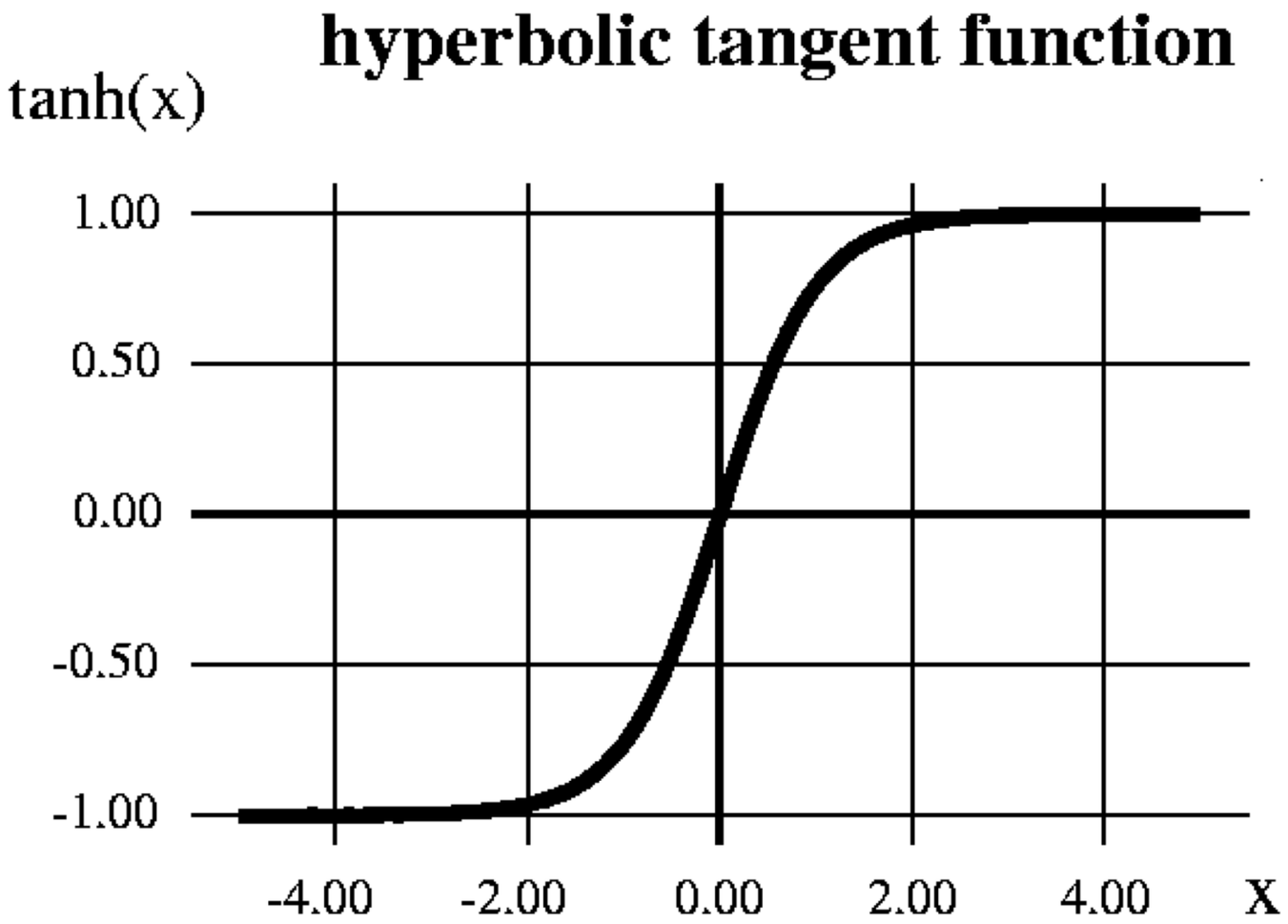
- When fitting a model, we:
  - Identify some loss function to optimize.
  - Pick a first guess.
  - Take a step of fixed size in the “best” direction.
  - Keep going until we’ve found the minimum of our loss function!

POTENTIAL PITFALLS



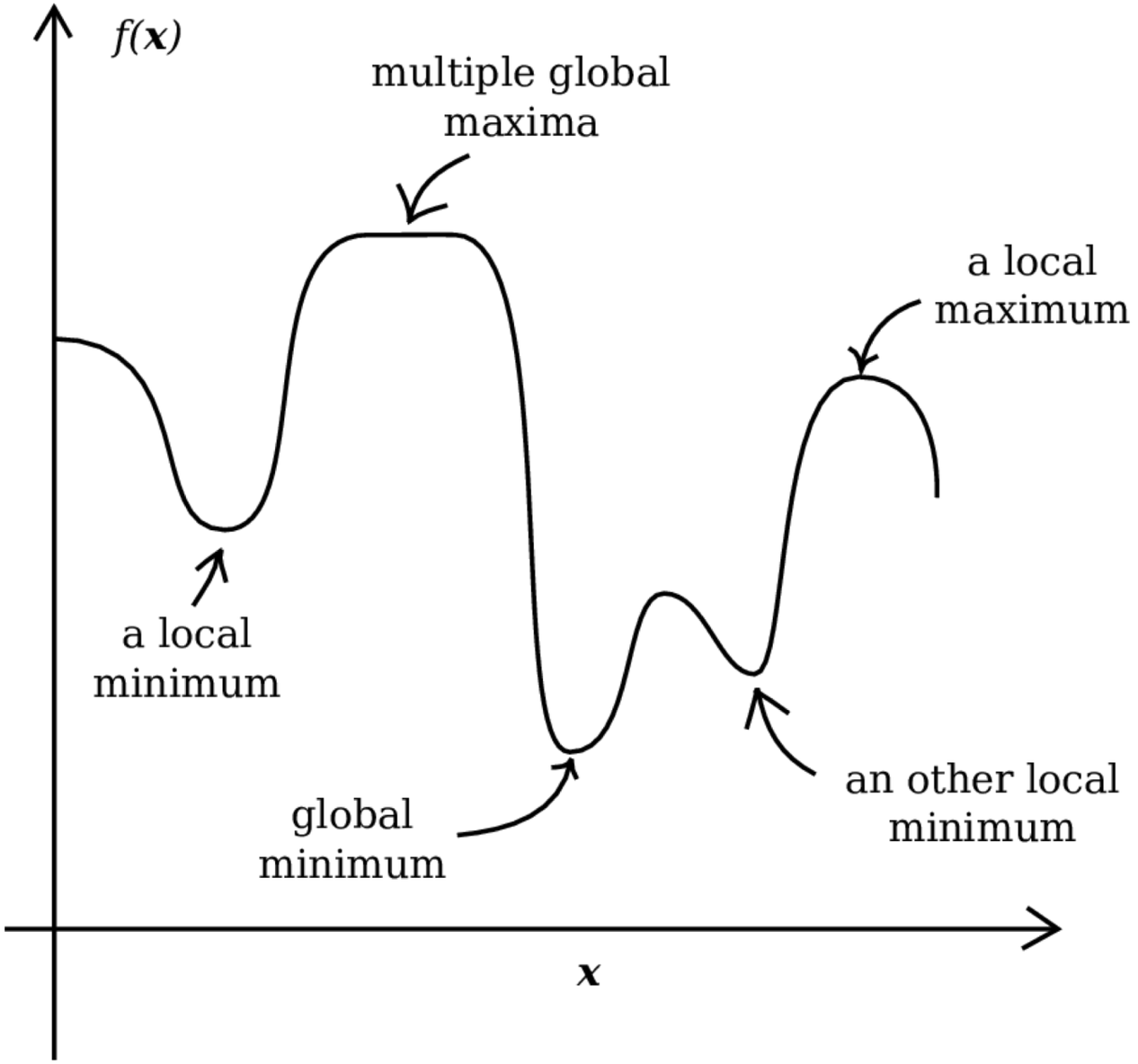
- If our step size is too big, we may never converge!
- If our step size is too small, it may take a very long time for us to converge!

POTENTIAL PITFALLS



- Depending on the shape of the loss function, gradient (slope of the curve) may be close to zero, meaning that learning occurs very slowly.

POTENTIAL PITFALLS



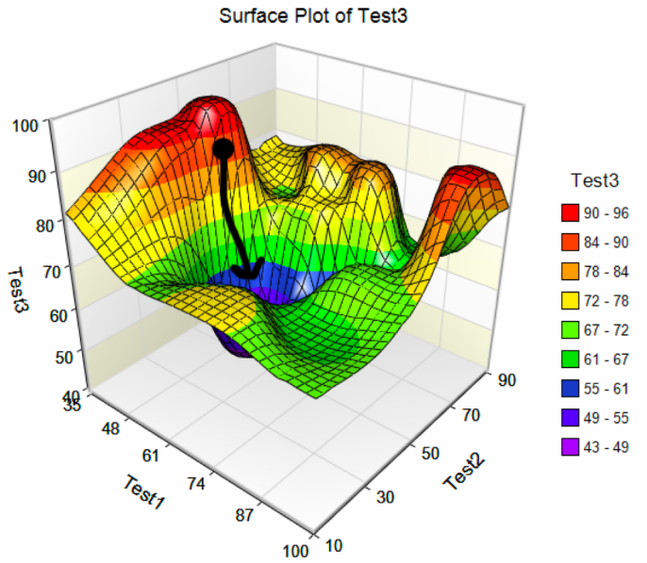
- Depending on the shape of the loss function, we may converge to a local optimum.
- This is one reason we attempt to choose convex loss functions.
- Shockingly, this isn't a major problem.

SOLUTION 1: ADAPTIVE GRADIENT DESCENT

- One way to attempt to protect against some of these pitfalls is to use **adaptive gradient descent**, which means that, at each step, we draw α from some distribution so that we aren't taking a step of fixed size.
  - There are many different flavors of "adaptive" gradient descent algorithms for various purposes.

SOLUTION 2: CHANGE STARTING POINTS

- Another way to attempt to protect against some of these pitfalls is to change the starting points of the algorithm.



- Intuitively, if we get nearly identical results a few, we can be more confident that we've arrived at the global optimum.

MACHINE LEARNING = GRADIENT DESCENT



Tim Hopper
@tdhopper

Machine learning path to fame:
1) Find learning algorithm requiring optimization
2) Apply gradient descent
3) Give this algorithm cool name

